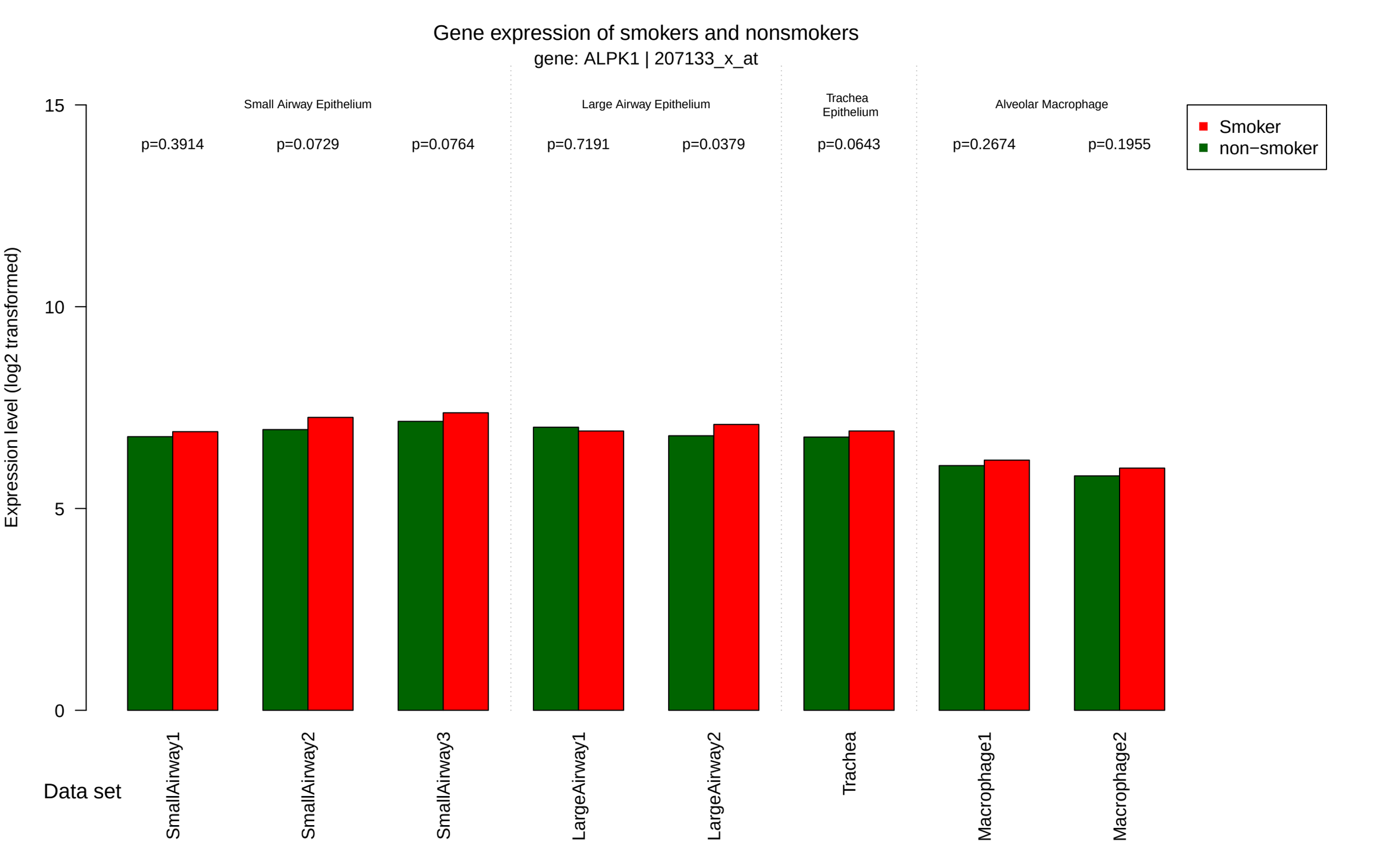
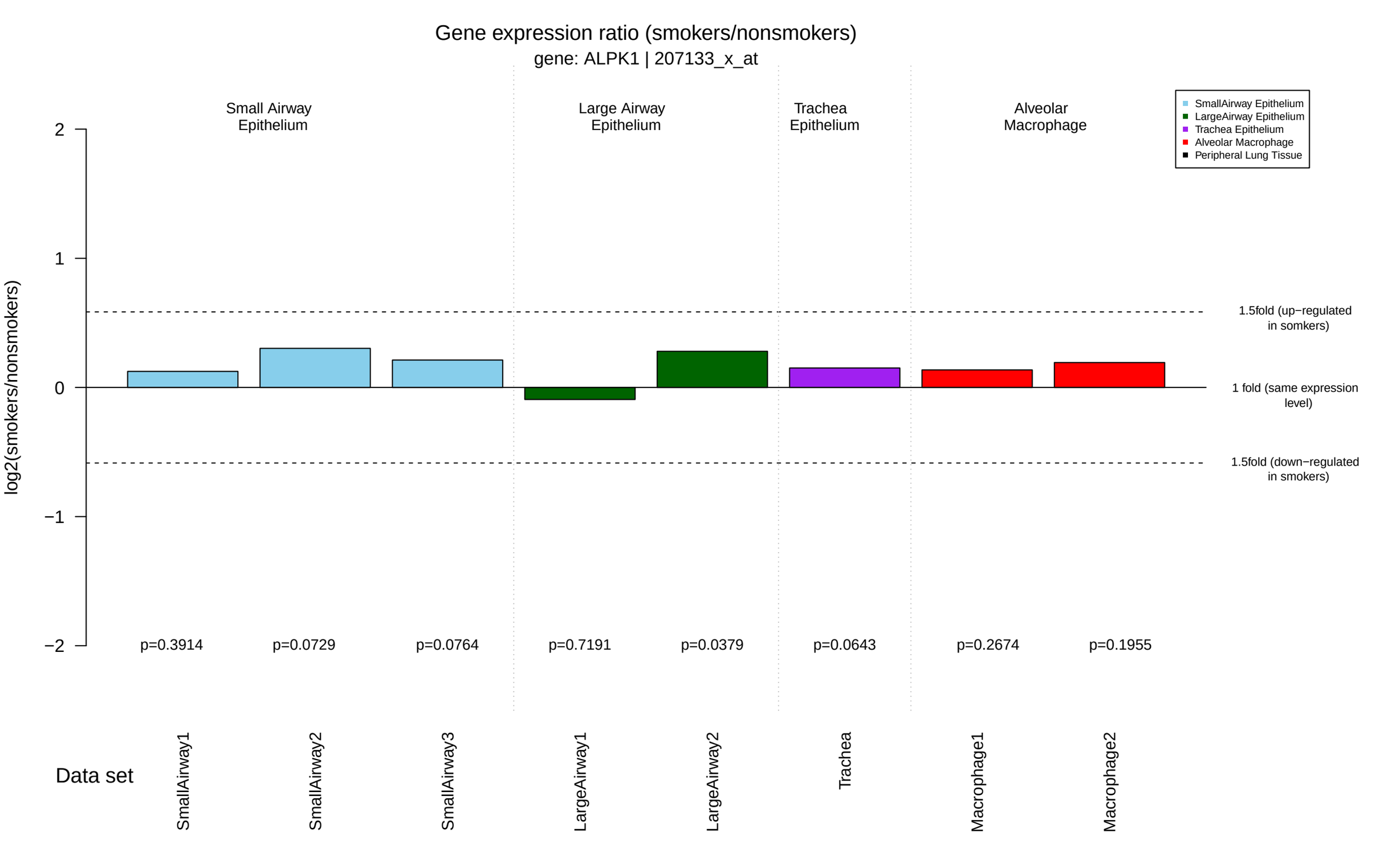
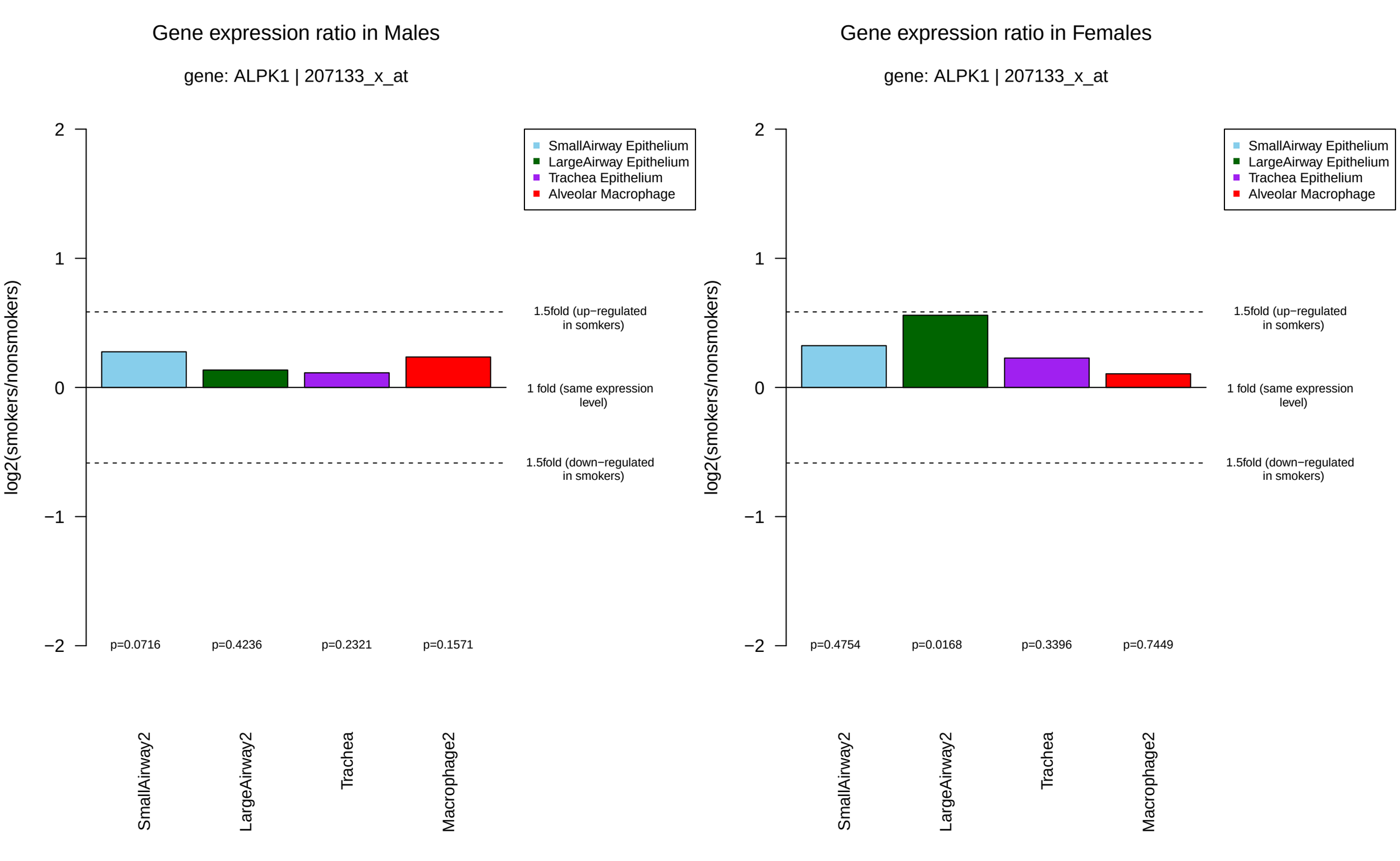
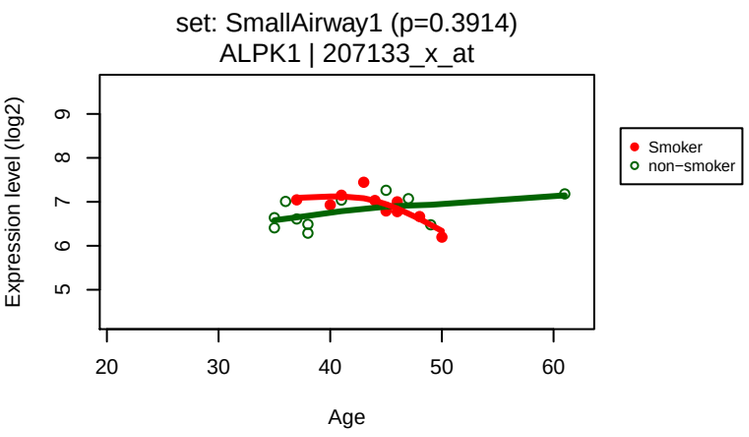
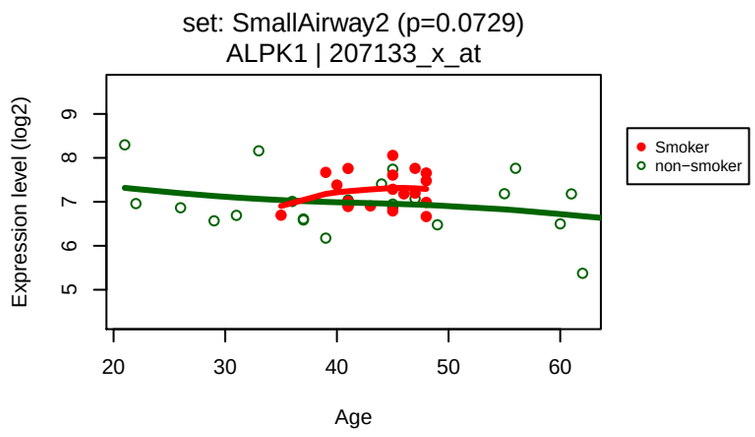
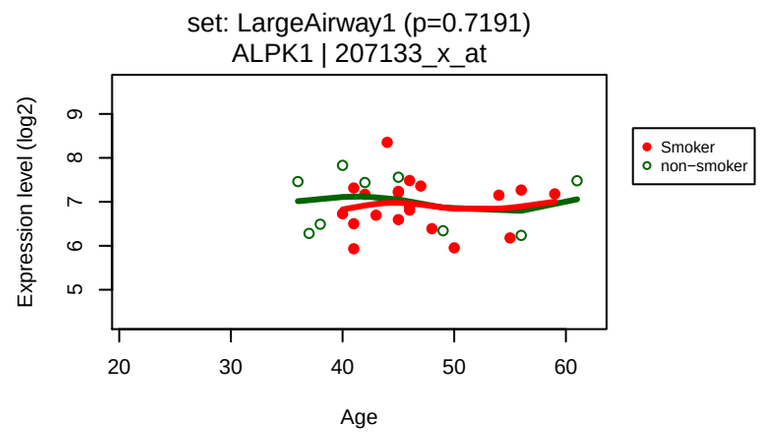

Gene expression of smokers and nonsmokers
gene: ALPK1 | 207133_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3914
p=0.0729
p=0.0764
p=0.7191
p=0.0379
p=0.0643
p=0.2674
p=0.1955
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

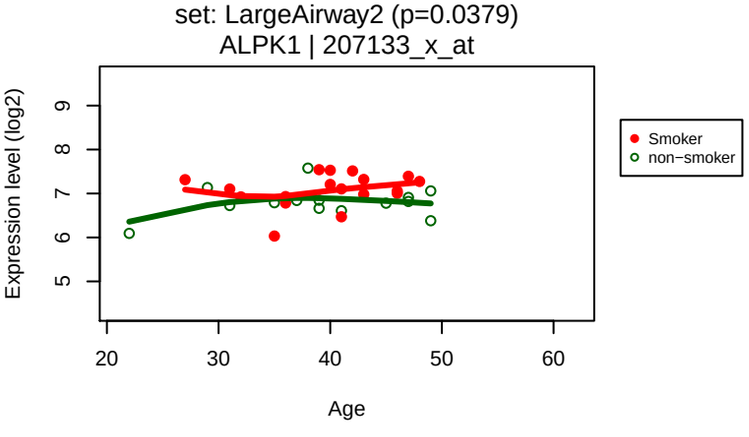

Gene expression ratio (smokers/nonsmokers)
gene: ALPK1 | 207133_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3914
p=0.0729
p=0.0764
p=0.7191
p=0.0379
p=0.0643
p=0.2674
p=0.1955
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

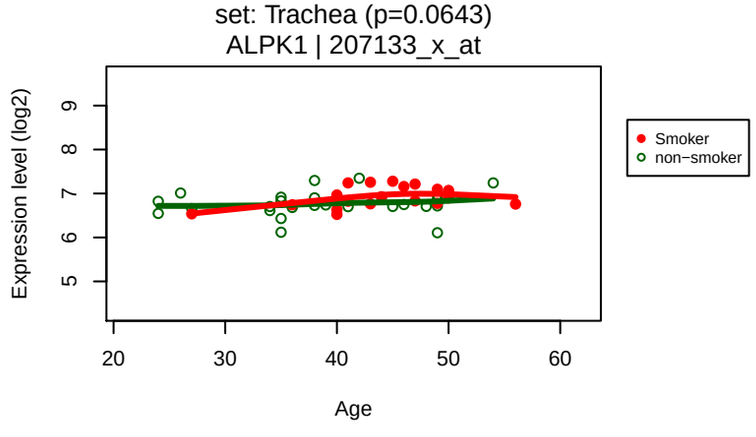

Gene expression ratio in Males
gene: ALPK1 | 207133_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0716
p=0.4236
p=0.2321
p=0.1571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALPK1 | 207133_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4754
p=0.0168
p=0.3396
p=0.7449
SmallAirway2
LargeAirway2
Trachea
Macrophage2

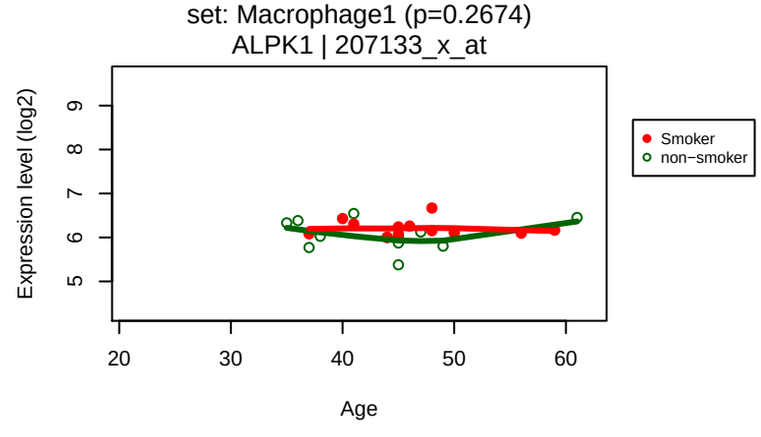

set: SmallAirway1 (p=0.3914)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker

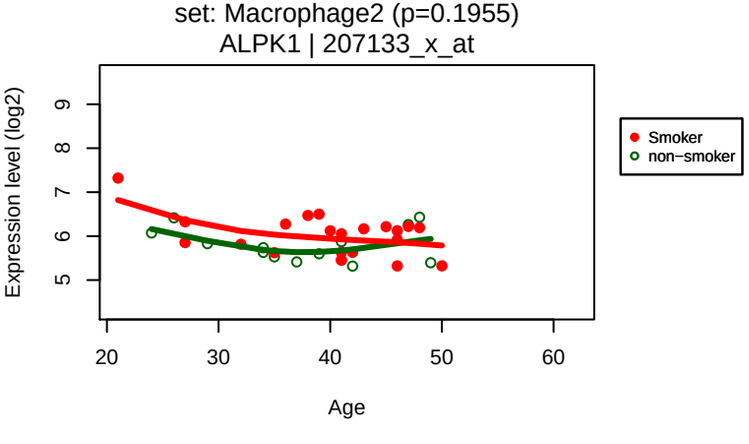

set: SmallAirway2 (p=0.0729)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker

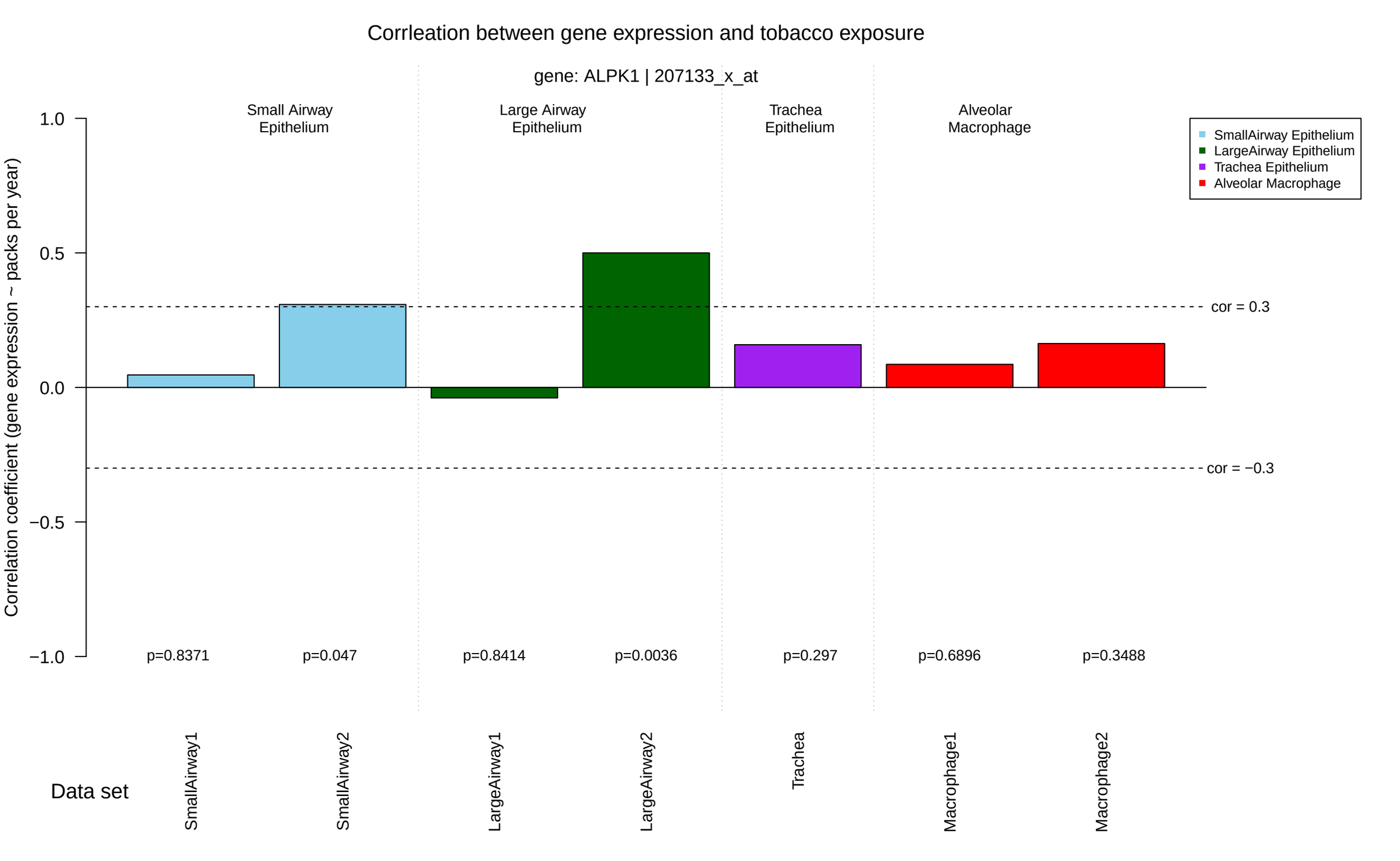

set: LargeAirway1 (p=0.7191)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0379)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0643)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2674)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1955)
ALPK1 | 207133_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALPK1 | 207133_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8371
p=0.047
p=0.8414
p=0.0036
p=0.297
p=0.6896
p=0.3488
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2